

Full Stack Developer Assignment

Assignment: Design & Develop Meta Accounts Center

Objective:

Build a simplified version of Meta Accounts Center, where users can securely manage their connected accounts, personal information, login settings, and privacy preferences.

The goal of this assignment is to evaluate your ability to design scalable applications, write clean code, build secure APIs, and create an intuitive user experience.

Problem Statement:

Design and develop a web application that replicates the core functionality of Meta Accounts Center.

The application should allow users to:

- Register and log in securely.
- View and edit their profile information.
- Connect multiple social accounts.
- Manage security settings.
- Configure privacy preferences.
- View recent account activities.
- Manage connected devices.

The application should demonstrate good architecture, responsiveness, security practices, and code quality.

Functional Requirements

1. Authentication

Implement:

- User Registration
- Login
- Logout
- Forgot Password (mock implementation is acceptable)
- JWT or Session-based Authentication

2. Dashboard

After login, display an overview containing:

- User Profile
- Connected Accounts
- Security Status
- Recent Activities
- Connected Devices

3. Profile Management

Allow users to:

- Update Name
- Update Email
- Update Phone Number
- Upload Profile Picture (optional)
- Update Date of Birth

4. Connected Accounts

Users should be able to:

- View connected accounts
- Connect new accounts (mock providers such as Facebook, Instagram, WhatsApp)
- Remove connected accounts

No real OAuth integration is required. Mock implementations are acceptable.

5. Security Center

Implement:

- Change Password
- Enable/Disable Two-Factor Authentication (mock)
- Display active login sessions
- Logout from individual sessions
- Logout from all devices

6. Privacy Settings

Allow users to configure:

- Profile Visibility
- Email Visibility
- Phone Number Visibility
- Personalized Ads Preference
- Data Sharing Preference

7. Activity History

Display:

- Login history
- Password changes
- Profile updates
- Connected account changes
- Privacy setting changes

Include:

- Date & Time
- Action Type
- Device Name
- Browser
- IP Address (mock)

8. Device Management

Users should be able to:

- View active devices
- View browser details
- View operating system
- View login time
- Remove a device session

Backend Requirements

Develop REST APIs for:

- Authentication
- User Profile
- Connected Accounts
- Security
- Privacy Settings
- Activity Logs
- Device Management

Implement proper:

- Validation
- Error Handling
- Authentication Middleware
- Authorization
- Status Codes

Database

Design a normalized database for:

- Users
- Connected Accounts
- User Sessions
- Activity Logs
- Privacy Settings
- Security Settings

You may use:

- PostgreSQL
- MySQL
- MongoDB

Frontend Requirements

The application should:

- Be fully responsive
- Match modern UI standards
- Provide smooth navigation
- Include loading states
- Include validation messages
- Display empty states where applicable

Use any modern framework such as:

- React
- Next.js
- Vue
- Angular

Technical Expectations

Your project should demonstrate:

- Clean folder structure
- Reusable components
- Modular backend architecture
- Proper state management
- Secure authentication
- RESTful API design
- Clean coding practices
- Proper Git commit history

Bonus Features

Additional credit will be given for implementing:

- Dark Mode
- Role-Based Access Control
- Email Verification
- Unit Testing
- Docker Support
- API Documentation (Swagger/OpenAPI)
- CI/CD Pipeline
- Rate Limiting
- Caching
- Optimistic UI Updates

Deliverables

Please submit:

- Source code (GitHub repository)
- README with setup instructions
- Database schema or ER diagram
- API documentation
- Environment variables sample (.env.example)
- Screenshots or short demo video (optional)

NOTE: The use of AI tools (such as ChatGPT, GitHub Copilot, Claude, Gemini, Cursor, or similar assistants) is **permitted** for this assignment. However, candidates are expected to fully understand and be able to explain every part of the code they submit.

During the technical evaluation or interview, you may be asked to discuss your implementation, architectural decisions, and specific code segments. Any submission found to be copied without sufficient understanding may negatively impact the evaluation.



Please ensure that all code submitted is your own work, even if AI tools were used to assist in development.

Submission Guidelines:

- Complete the assignment within 2days.
- Share it to sanket.debnath@redsoftware.in, keeping hr.redsoftware.in the CC.
- Upload the project to a GitHub repository.
- Ensure the repository includes a detailed README with installation and execution steps.
- Include any assumptions or trade-offs made during development.
- Share the GitHub repository link along with any additional documentation before the submission deadline.

Good luck! We look forward to reviewing your solution.